



Shahid Beheshti University

Department of Computer Science
Advanced Data Mining / Deep Learning

Mini Project 3

TF-1: Transformer Text Classification

Fine-tuning DistilBERT on AG News

Project Information

Architecture family:	Transformer
Dataset:	AG News
Project value:	25 points
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Head TA:	Mr. Mobin Nesari
Student:	Alireza Jahanbakhsh
Date:	August 17, 2026
Notebook Link:	Kaggle Notebook
Project Link:	GitHub Repository

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Data Mining Course

1 Problem and Data

This project addresses four-class topic classification on AG News. The executed notebook loaded the Hugging Face `ag_news` dataset, containing 120,000 official training examples and 7,600 official test examples. The label mapping used in the notebook was 0: World, 1: Sports, 2: Business, and 3: Sci/Tech.

The official test set was kept untouched until the final fixed-budget comparison. The original training split was divided with stratification and seed 42 into a 108,000-example training pool and a 12,000-example validation set. For the original contribution, balanced stratified subsets of 500, 2,000, and 8,000 labeled examples were drawn from the training pool. These subsets contained 125, 500, and 2,000 examples per class, respectively.

The two model families used different input representations. The classical baseline lowercased text, removed English stop words, and constructed TF-IDF unigram and bigram features. The Transformer used the `distilbert-base-uncased` tokenizer with truncation and padding to a maximum sequence length of 128 tokens. Validation data were used for the data-efficiency study, checkpoint selection within each DistilBERT run, and temperature selection in the calibration stretch experiment.

2 Method

2.1 TF-IDF + Logistic Regression Baseline

The required classical baseline was TF-IDF followed by Logistic Regression. The vectorizer used lowercasing, English stop-word removal, `ngram_range=(1,2)`, `min_df=2`, and at most 50,000 features. Logistic Regression used `max_iter=1000`, no class weighting, `n_jobs=-1`, and random seed 42.

2.2 DistilBERT Fine-Tuning

The Transformer backbone was `distilbert-base-uncased` with a four-class sequence-classification head. The notebook fine-tuned the model with AdamW at learning rate 2×10^{-5} , batch size 16, and two epochs. After each epoch, validation Macro-F1 was computed and the state with the highest validation Macro-F1 was restored. Training ran on CUDA in the executed notebook.

Evaluation used Accuracy and Macro-F1 for the data-efficiency and final comparisons. Per-class Precision, Recall, and F1 were additionally reported on the final test set. Macro-F1 is especially useful here because it gives the four classes equal contribution to the summary score.

3 Original Contribution

3.1 Hypothesis

The notebook states the following hypothesis:

A pretrained contextual Transformer should preserve stronger macro-F1 than TF-IDF when labeled training data is limited, and should beat the classical baseline at the largest training budget.

The original contribution was therefore a controlled **data-efficiency study**, not merely a single DistilBERT fine-tuning run. TF-IDF and DistilBERT were trained on the same stratified labeled-data budgets of 500, 2,000, and 8,000 examples and evaluated on the same validation split.

3.2 Validation Evidence

The validation results are shown in Table 1. DistilBERT achieved higher Macro-F1 at all three data budgets. The Macro-F1 gaps, DistilBERT minus TF-IDF, were 0.0918, 0.0395, and 0.0170 for 500, 2,000, and 8,000 examples, respectively. The notebook reported a mean low-data gap of 0.0656 over the 500- and 2,000-example budgets and marked the validation hypothesis as **SUPPORTED**. The result is consistent with the stated hypothesis: the Transformer advantage was largest under limited labeled data and narrowed as the training set grew.

Table 1: Validation data-efficiency results from the executed notebook.

Method	Train size	Accuracy	Macro-F1	Training time (s)
DistilBERT	500	0.8748	0.8748	108.9138
TF-IDF + Logistic Regression	500	0.7839	0.7830	—
DistilBERT	2,000	0.9046	0.9043	148.4905
TF-IDF + Logistic Regression	2,000	0.8654	0.8648	—
DistilBERT	8,000	0.9126	0.9128	293.0329
TF-IDF + Logistic Regression	8,000	0.8962	0.8958	—

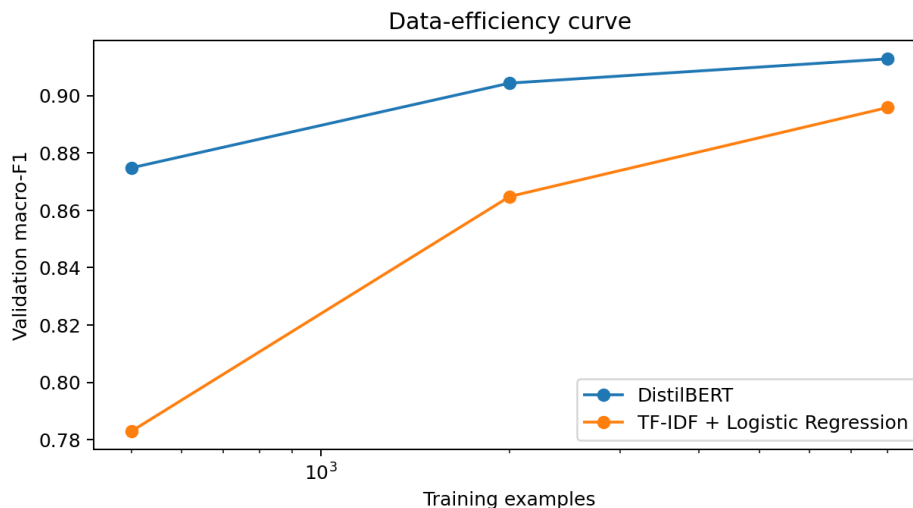


Figure 1: Validation Macro-F1 across the three labeled-data budgets. DistilBERT remained ahead at every budget, while the gap decreased as labeled data increased.

4 Experiments

The main experiment compared TF-IDF + Logistic Regression and DistilBERT at the three controlled labeled-data budgets. The final test comparison used the fixed 8,000-example budget, the largest budget in the study, after the validation data-efficiency analysis had been completed.

The notebook also contained the following supporting analyses:

- final per-class Precision, Recall, and F1;
- confusion matrices and representative model disagreements;
- short-text and confidence error profiling;
- an exact McNemar test on paired correctness outcomes;
- a paired bootstrap 95% confidence interval for the Macro-F1 difference;
- a calibration stretch experiment with Expected Calibration Error and temperature scaling;
- a controlled Text-CNN, self-attention, and CNN+self-attention experiment for the required Discussion & Observation questions.

5 Results and Analysis

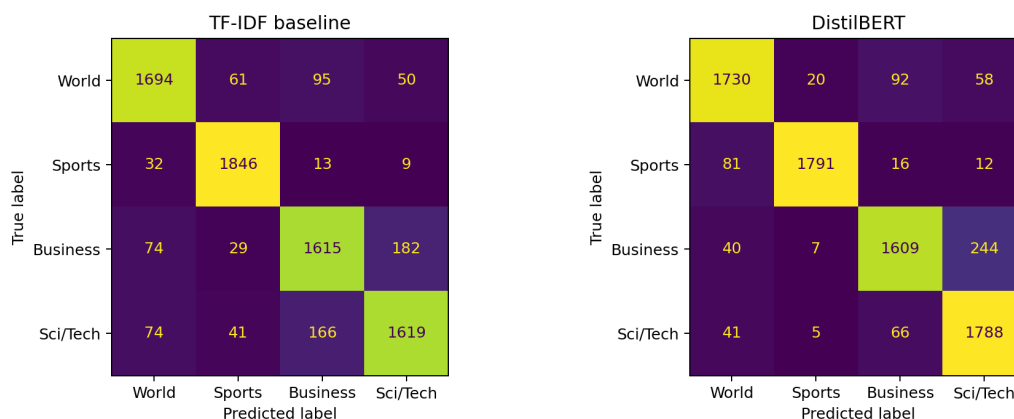
5.1 Final Test Comparison

At the final 8,000-example training budget, DistilBERT achieved test Accuracy 0.9103 and Macro-F1 0.9105, compared with 0.8913 Accuracy and 0.8909 Macro-F1 for TF-IDF + Logistic Regression.

Table 2: Final test comparison at the fixed 8,000-example training budget.

Model	Train size	Accuracy	Macro-F1
TF-IDF + Logistic Regression	8,000	0.8913	0.8909
DistilBERT	8,000	0.9103	0.9105

Because both models were evaluated on the same 7,600 test examples, the notebook used paired statistical analysis. DistilBERT was uniquely correct on 424 examples, whereas TF-IDF was uniquely correct on 280 examples, a net difference of 144 examples. The exact McNemar test was computed from these discordant outcomes; the notebook’s rounded result was reported as $p = 0.0$. The paired bootstrap estimate for the Macro-F1 difference was 0.01956 with a 95% confidence interval of $[0.01277, 0.02637]$. The interval excludes zero and supports a positive test-set Macro-F1 difference for DistilBERT under this experiment.

**Figure 2:** Final test confusion matrices for TF-IDF + Logistic Regression and DistilBERT.

5.2 Per-Class Results

DistilBERT improved F1 for all four classes. The largest F1 gain was on Sci/Tech and the smallest was on Sports. The full per-class comparison is reported in Table 3.

Table 3: Per-class test metrics from the executed notebook.

Class	TF-IDF P	TF-IDF R	TF-IDF F1	DistilBERT P	DistilBERT R	DistilBERT F1	$\Delta F1$
World	0.9039	0.8916	0.8977	0.9144	0.9105	0.9124	0.0147
Sports	0.9337	0.9716	0.9523	0.9824	0.9426	0.9621	0.0098
Business	0.8549	0.8500	0.8525	0.9024	0.8468	0.8737	0.0213
Sci/Tech	0.8704	0.8521	0.8612	0.8506	0.9411	0.8936	0.0324

For Sci/Tech, DistilBERT Recall increased from 0.8521 to 0.9411 while Precision decreased from 0.8704 to 0.8506. For Business, DistilBERT Precision increased from 0.8549 to 0.9024 while Recall changed from 0.8500 to 0.8468. These class-level results show that the aggregate gain was not produced by the same precision–recall behavior in every class.

5.3 Representative Errors and Error Profile

The notebook exported representative cases from both disagreement directions: DistilBERT wrong / TF-IDF correct and TF-IDF wrong / DistilBERT correct. For example, a PeopleSoft item with true label Business was predicted as Sci/Tech by DistilBERT while TF-IDF was correct. Conversely, an item about undelivered refund checks with true label Business was correctly classified by DistilBERT while TF-IDF predicted Sci/Tech. These cases demonstrate that the two models can disagree even when their overall performance is relatively close.

The DistilBERT test error profile contained 682 incorrect and 6,918 correct predictions. Mean text length was 38.403 words for incorrect predictions and 37.655 for correct predictions, with a median of 37 in both groups. Mean confidence was 0.807 for incorrect predictions and 0.957 for correct predictions. In this analysis, confidence separated correct and incorrect cases more clearly than text length.

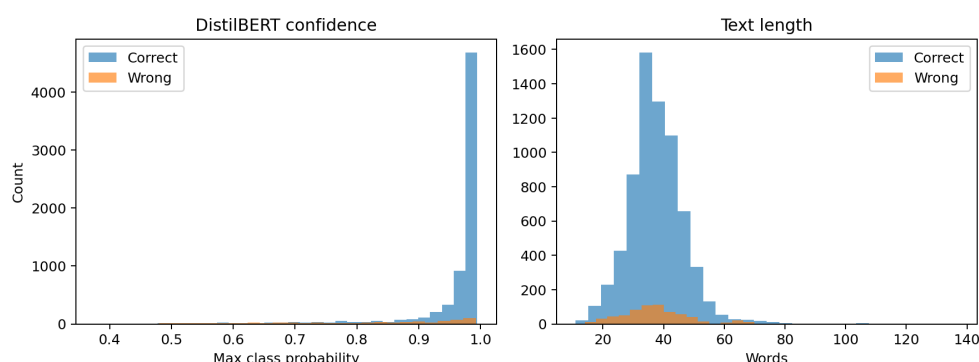


Figure 3: DistilBERT test error profile. Text length was similar for correct and incorrect cases, while mean confidence differed substantially.

5.4 Statistical Summary

The notebook’s statistical comparison combined the exact McNemar test with a 2,000-resample paired bootstrap for the Macro-F1 difference. McNemar’s test used the paired correctness outcomes, while the bootstrap resampled test examples jointly for both models. The observed Macro-F1 difference was 0.01956 and the bootstrap 95% interval was [0.01277, 0.02637].

6 Stretch Bonus: Calibration Analysis

The TF-1 project card lists calibration analysis as a stretch goal. The notebook implemented this extension without changing the trained classifier.

Expected Calibration Error (ECE) was computed with ten confidence bins. Before temperature scaling, TF-IDF + Logistic Regression had ECE 0.1878 and DistilBERT had ECE 0.0333. A positive temperature was selected using validation logits by minimizing negative log-likelihood. The selected value was $T = 1.2$. On the test set, DistilBERT ECE decreased

from 0.0333 to 0.0165, while negative log-likelihood decreased from 0.2747 to 0.2681. Because a positive temperature rescales all logits for a sample by the same factor, the predicted class ordering was unchanged; the intervention affected confidence calibration rather than classification accuracy.

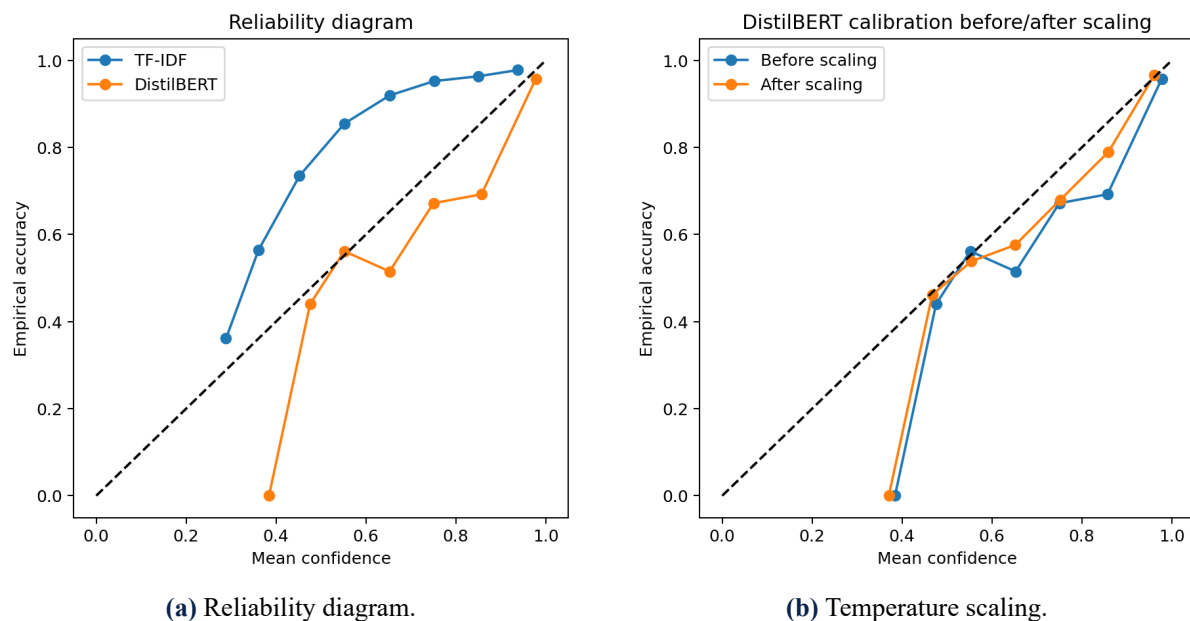


Figure 4: Calibration stretch experiment from the notebook.

7 Discussion and Observation

7.1 Question 1: Convolution versus Self-Attention

The handbook asks what a 1-D convolution can capture that self-attention does not encode in the same way, and vice versa. In the notebook's controlled supplementary experiment, all three lightweight models used the same 8,000-example subset and the same validation split. TextCNN achieved validation Macro-F1 0.8091, while the self-attention classifier achieved 0.8058. Their aggregate performance was therefore very similar.

The architectural difference is their inductive bias. A 1-D convolution directly emphasizes neighboring token patterns through a local kernel, whereas self-attention can combine evidence from positions that are far apart in the sequence. The notebook used an actual validation disagreement to ground this comparison:

Intel Unwires Chinas Panda Playground In Wolong Sydney, Australia, 25 October, 2004 - Intel Corporation today announced it has deployed a campus-wide broadband and wireless communications network within Chinas Wolong Nature Reserve to aid researchers in the study, care and preservation of the

The true class was Sci/Tech. Text-CNN predicted World, while the self-attention model predicted Sci/Tech. Together with the similar overall Macro-F1 scores, this example supports the notebook's conclusion that neither mechanism was uniformly superior; they sometimes relied on different evidence.

7.2 Question 2: Hybrid Convolution and Attention

The notebook explicitly tested the requested hybrid:

Embedding → 1-D Convolution → Self-Attention → Masked Pooling → Classifier.

Attention was placed after convolution so that local phrase-level features were formed first and then combined across longer distances. The controlled comparison used the same tokenizer, embedding dimension, training subset, optimizer family, epoch budget, and validation data.

Table 4: Controlled supplementary experiment for the Discussion & Observation questions.

Model	Accuracy	Macro-F1	Parameters	Training time (s)
Text-CNN	0.8093	0.8091	3,956,868	3.0678
Self-attention	0.8072	0.8058	4,056,196	4.8474
CNN + self-attention	0.8097	0.8094	4,039,556	4.6784

The hybrid had the highest validation Macro-F1, but its advantage over Text-CNN was only 0.0003. Therefore, this experiment does not establish a large performance gain from combining the two mechanisms. It does show that the design can combine local convolutional features with longer-range self-attention without a meaningful degradation under this compact setup.

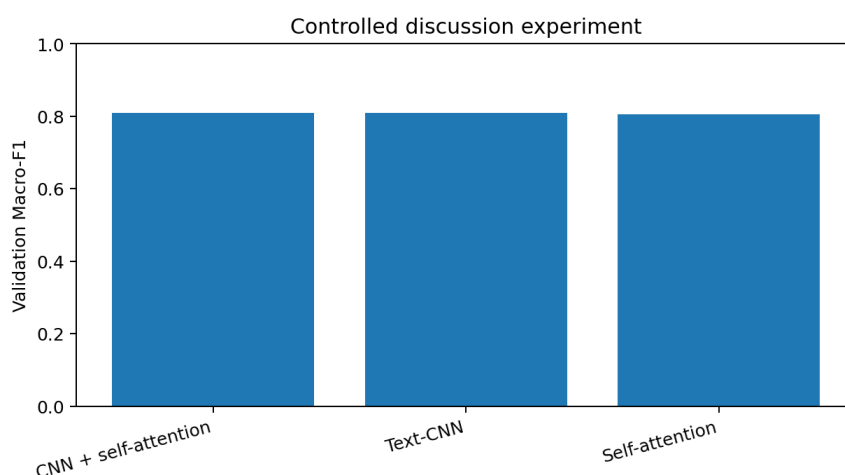


Figure 5: Validation Macro-F1 for Text-CNN, self-attention, and the CNN+self-attention hybrid.

8 Limitations and Next Steps

The data-efficiency study used one random seed and only three labeled-data budgets. The comparison controls the amount of task-specific labeled data, but it does not control pretraining compute because DistilBERT begins from pretrained representations whereas TF-IDF + Logistic Regression does not. DistilBERT fine-tuning was also intentionally compact, with two epochs per budget.

The representative error cases are examples rather than a complete error taxonomy. The calibration result depends on the ECE binning scheme, and temperature scaling was tuned on the single validation split. The controlled CNN/self-attention/hybrid experiment was evaluated only on validation data and used a short three-epoch budget.

Useful next steps would be to repeat the data-efficiency experiment over multiple seeds, introduce additional labeled-data budgets, evaluate the handbook’s alternative robustness-to-distribution-shift stretch option, and test whether the supplementary CNN/self-attention conclusions persist under longer training and final-test evaluation.